

MAGNETOM

Installation and Startup MAGNETOM Options

Upgrade Tim

Avanto
Espree
Trio, A Tim System
Symphony, A Tim System
Verio

Document Version

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1.1 Safety Information

For your safety and the safety of third parties as well as for protecting the MR system and its components from damage, the appropriate safety information is included in register "General Safety Notes" and "MR-Specific Safety Information" in the "Technical Documents".

Prior to starting work, read the specific information in the respective manual:

- General safety information for service personnel.
- Limitations with respect to ferromagnetic materials.
- Safety information pertaining to working with the magnet at field.
- Information regarding working with heavy loads.
- Protective conductor test before switching on the system and for subsequent repairs.

1.2 General information

The enclosed instructions are intended for the following upgrade packages:

Avanto	Upgrade package	GPA upgrade
from 8 to 18 channels	Upgr. Tim (32x8)Q - (76x18)Q	n.a.
from 8 to 18 channels	Upgr. Tim (32x8)Q - (76x18)SQ	Q to SQ engine
from 8 to 32 channels	Upgr. Tim (32x8)Q - (76x32)SQ	Q to SQ engine
from 18 to 32 channels	Upgr. Tim (76x18)Q - (76x32)SQ	Q to SQ engine
from 18 to 32 channels	Upgr. Tim (76x18)SQ - (76x32)SQ	n.a.
18 channels	Upgr. Tim (76x18)Q - (76x18)SQ	Q to SQ engine

Espreo	Update Kit	GPA upgrade
from 8 to 18 channels	Upgr. Tim (32x8)Z - (76x18)Z	n.a.
from 8 to 18 channels	Upgr. Tim (32x8)Z - (76x18)DZ	Z to DZ engine ¹
from 8 to 32 channels	Upgr. Tim (32x8)Z - (76x32)DZ	Z to DZ engine ¹
from 18 to 32 channels	Upgr. Tim (76x18)Z - (76x32)DZ	Z to DZ engine ¹
from 18 to 32 channels	Upgr. Tim (76x18)Z - (76x32)Z	n.a.
18 channels	Upgr. Tim (76x18)Z - (76x18)DZ	Z to DZ engine ¹

1. The **slimline gradient coil** is a prerequisite for the upgrade from Z to DZ.



The "Upgrade kit, divided rear cover" (material number 10018727) is necessary for Espreo systems with **serial numbers 30096** and lower.

Symphony A Tim System	Upgrade package	GPA upgrade
from 18 to 32 channels	Upgr. Tim (76x18) - (76x32)	n.a.

Trio A Tim System	Upgrade package	GPA upgrade
from 8 to 18 channels	Upgr. Tim (102x8) - (102x18)	n.a.
from 8 to 32 channels	Upgr. Tim (102x8) - (102x32)	n.a.
from 18 to 32 channels	Upgr. Tim (102x18) - (102x32)	n.a.

Verio System:	Upgrade package	GPA upgrade
from 8 to 18 channels	Upgr. Tim (102x8) - (102x18)	n.a.

Verio System:	Upgrade package	GPA upgrade
from 8 to 32 channels	Upgr. Tim (102x8) - (102x32)	n.a.
from 18 to 32 channels	Upgr. Tim (102x18) - (102x32)	n.a.



The instructions for a GPA Upgrade only are included in the Start-Up (software installation).



The instructions show the upgrade from 8 to 32 channels.

1.2.1 Requirements

Espreo System:

The **slimline gradient coil** is a prerequisite for upgrading from the Z to the DZ engine.

For systems up to **serial no. 30096**, the **back funnel** needs to be replaced (the new back funnel is in separate parts, with an insertion funnel), for more information refer to (→ Espreo covers / M6-020.841).

In systems with a **straight gradient coil**, this must be replaced by a **slim-line gradient coil**, for more information, refer to (→ Gradient coil / M6-020.841.16).

1.2.2 Required Tools and Auxiliary Devices

- Standard tool kit
- Anti-magnetic tool set
- ESD Kit
- Socket wrench for the RF cables in the ACC (is part of the delivery)
- 2 mm Allen key (for removing the blind plates and after inserting the PCI-TX boards)
- Torx key TX8 (for removing the blind plates and after inserting the REC assemblies)

1.2.3 Time required



Depending on the various numbers of RF channels, different expenditures are required during the RF upgrade.

Duration of the upgrade:

HR	approx. 4-8 hours ¹
----	--------------------------------

Duration of the upgrade:	
GPA	approx. 1/2 hour
Start-Up	approx. 3 1/2 hours

1. When it is difficult to access the cable duct, more time is required for routing the RF cables.



During the RF upgrade, a second person is required for cable routing from the magnet to the ACC.

1.2.4 Delivery volume



Check the parts using the material list supplied with the delivery volume.

- Licenses

Upgrade variations			
from 8 to 18 channels	2 PCI-RX ¹	2 Receivers	Cable + FOC + RF feedthrough
from 18 to 32 channels	1 PCI-RX ¹	1 Receivers	Cable + FOC + RF feedthrough
from 8 to 32 channels	3 PCI-RX ₁	3 Receivers	Cable + FOC + RF feedthrough

1. The PCI-RX boards are to be installed only for the MR00r Supermicro imager and MR005 Dual Opteron imager. The MCPPro Multi 2 and high-end imager are delivered new (replaced), and already equipped with the PCI-RX boards.

The upgrade from 8 to 32 channels comprises 2 kits: From 8 to 18 channels and from 18 to 32 channels.



There are different types of receivers: For more information, please refer to the SPC.

1.3 Workflow



EGB/ESD Kit has to be used when working with electronics.

- Check the parts using the material list supplied with the delivery volume.
- Checking the licenses (refer to the Start-up)
- Routing the cables in the ACC
- Routing the cable magnet-filter plate-ACC
- Removing the Imager and upgrading it
- Upgrading the RFSU
- Pulling in the fiber optic cable
- Installing the Imager
- Installing the licenses
- Installing the software Imager
- System configuration
- Tune-up
- Performing the test tools (MOD/REC test, functional RCCS test)
- QA
- Application check/customer demo

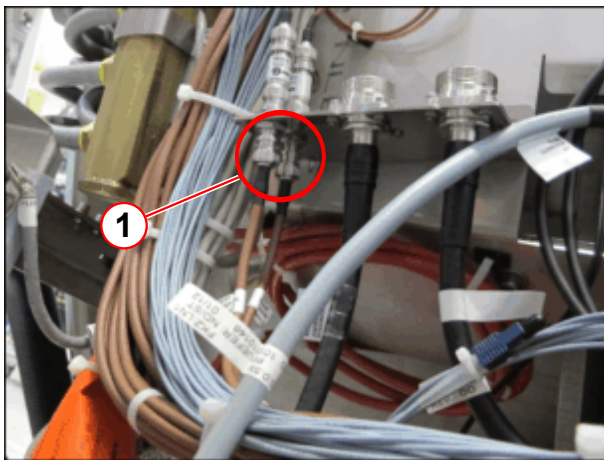
2.1 Routing the cables

- Ending Numaris (shutting down the software, no new start)
- Switch off the system.
- Switch off circuit breaker F100 in the mains box on top of the ACC.
- Test that voltage is no longer present

2.1.1 BC pickup coil connectors (only for Verio)

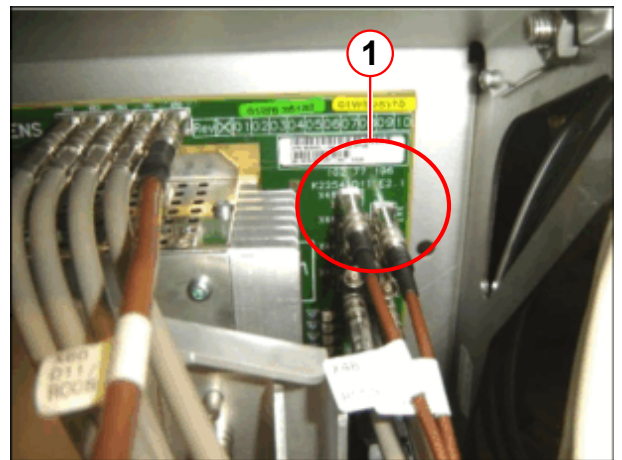
- Remove the RF cables W5185 and W5186 from the mounting plate connectors X10 and X11 and from RCCS X47 and X48 (they are no longer needed).

Fig. 1: 2 RF cables, X10 and X11



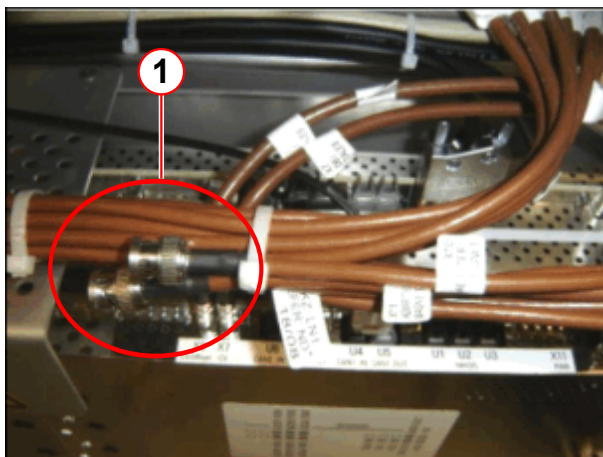
(1) X10 and X11

Fig. 2: RCCS X47 and X48



(1) X47 and X48

Fig. 3: X10 and X11 for 18 or 32 channel



(1) X10 and X11

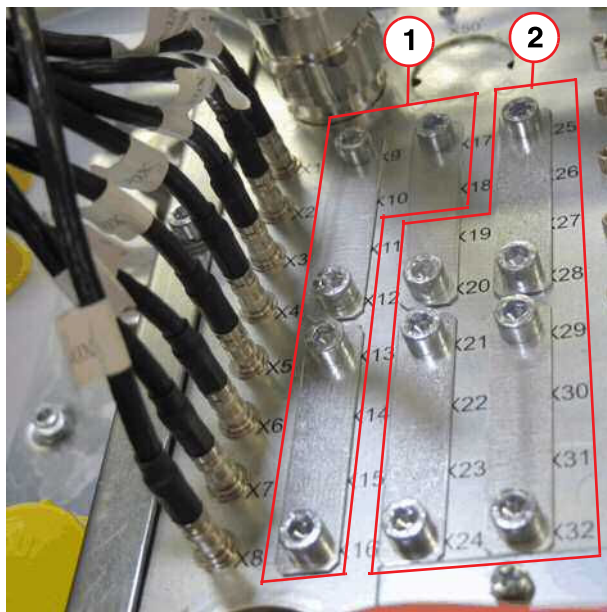
- Connect the two existing RF cables to mounting plate connectors X10 and X11.

2.1.2 Installing the RF cable in the ACC

For the **Avanto System**, the following steps must always be performed in the ACC cabinet, beginning with revision level **29**. For lower revision levels, the RF cables are already installed in the ACC cabinet. In this case, continue with (→ Cabling from the magnet to the ACC / Page 14).

For the **Espreo System**, **Symphony**, **A Tim System**, the **Trio**, **A Tim System**, and the **Verio System** the following steps must always be performed.

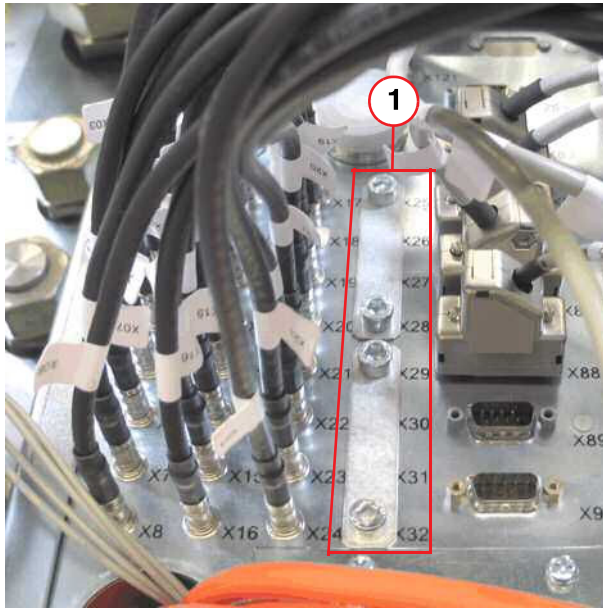
Fig. 4: Top of ACC



The picture to the left shows the position of the RF cables (QLA feed-throughs) for upgrades to 18 and/or 32 channels.

- (1) Upgrade 8 to 18 channels (X9 - X18)
- (2) Upgrade 18 to 32 channels (X19- X32)

Fig. 5: Top of ACC, old version with 18 channels (24 connections)

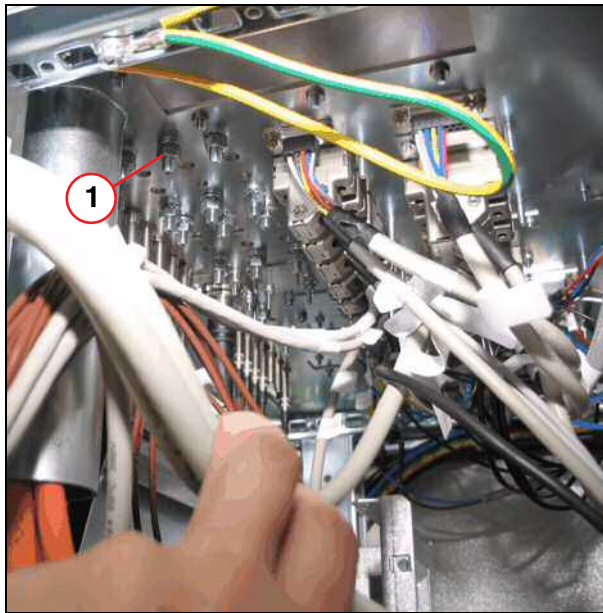


(1) Upgrade 18 to 32 channels (X25 - X32)

For older systems equipped with 18 channels and upgraded to 32 channels, the system is already equipped with 24 channels.

For the upgrade to 32 channels, only connectors X25 - X32 are additionally installed.

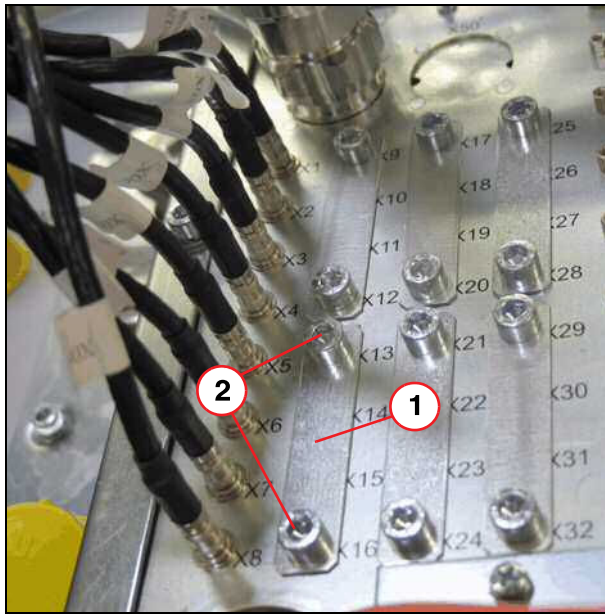
Fig. 6: ACC top plate, inside



(1) Nut, lock washer, washer

- Remove the nuts (1) depending on the revision level (→ Fig. 4 Page 10).

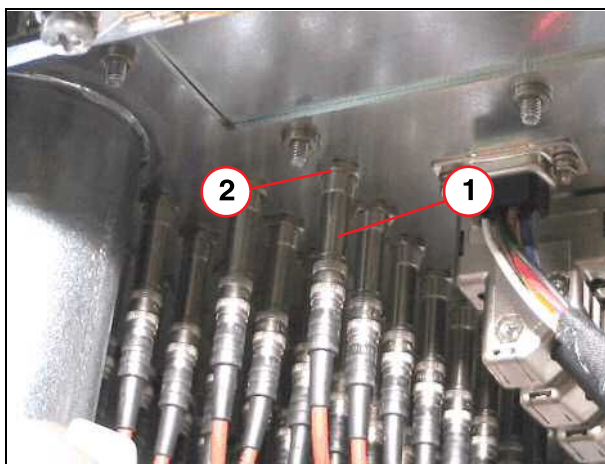
Fig. 7: ACC top plate



- (1) Cover plate
- (2) Screws

- Remove the screws (2) and covers (1) depending on the revision level (→ Fig. 4 Page 10).

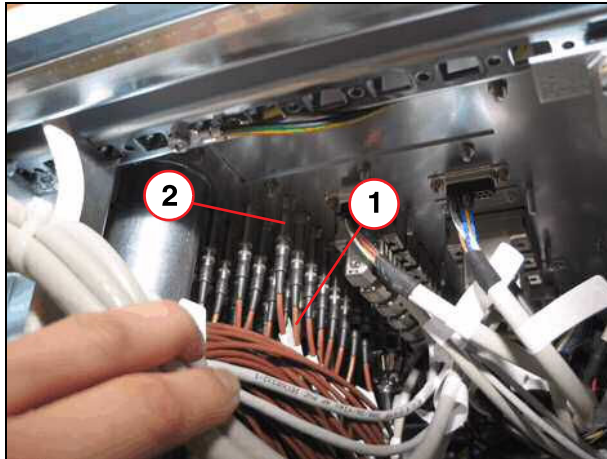
Fig. 8: ACC top plate, inside with RF cable



- (1) QLA adapter
- (2) Nut

- Inserting the QLA adapter (1) from above and securing it with nut (2) by using the socket wrench included in the delivery volume.

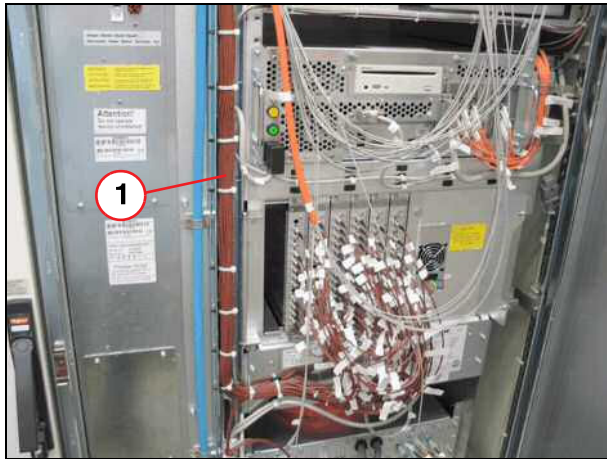
Fig. 9: ACC top plate, inside with RF cable



- (1) RF cable
- (2) QLA adapter

- Inserting the RF cables (1) according to the designations on the QLA adapters (1).

Fig. 10: Cable routing inside the ACC

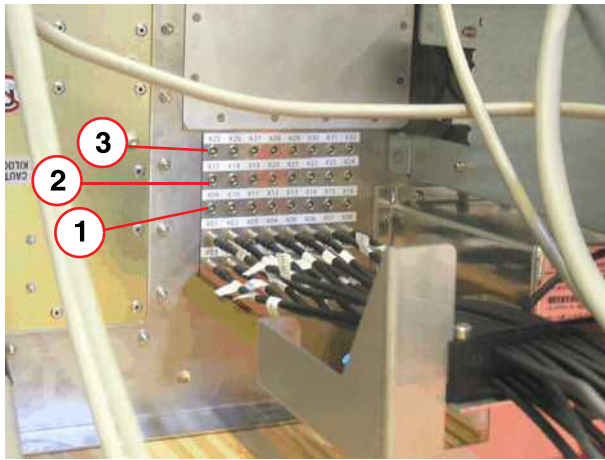


- (1) RF cable

- Routing the RF cables (1) from the top plate to the receivers.

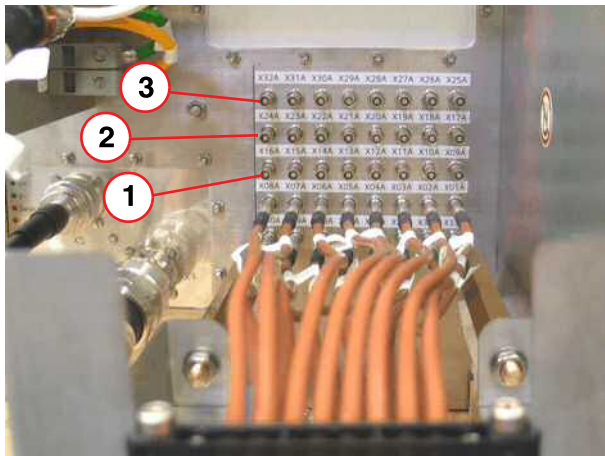
2.1.3 QLA feedthroughs on the filter plate (optional)

Fig. 11: QLA feedthrough (view - technical room)



- (1) X09 - X16
- (2) X17 - X24
- (3) X25 - X32

Fig. 12: QLA feedthrough (view - magnet room)



- (1) X09A - X16A
- (2) X17A - X24A
- (3) X25A - X32A

In newer systems, the QLA feedthroughs also have to be installed at the filter plate.

(2 reserve QLA feedthroughs are included in the delivery volume)

- Remove the screws and nuts at the filter plate:
 - 8 to 18 channels: **X9 - X18**
 - 18 to 32 channels: **X19 - X32**
 - 8 to 32 channels: **X9 - X32**
- Install the QLA feedthroughs at the filter plate:
 - 8 to 18 channels: **X9 - X18**
 - 18 to 32 channels: **X19 - X32**
 - 8 to 32 channels: **X9 - X32**



In older systems, the QLA feedthroughs are already installed on the filter plate; the QLA feedthroughs included in the delivery volume are not required.

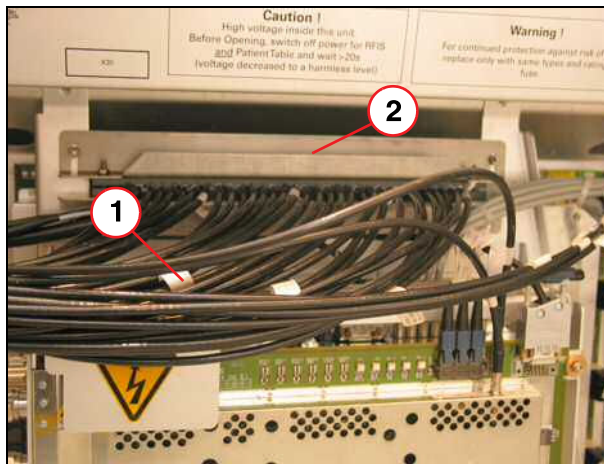
2.1.4 Cabling from the magnet to the ACC



In older systems (18 channel), the RF cables for 24 channels (X01 - X24) are already connected. The additional RF cables (X19-X24) are not required (needed for newer systems only).

Connecting and routing the cables from the magnet to the filter plate, and from the filter plate to the ACC cabinet.

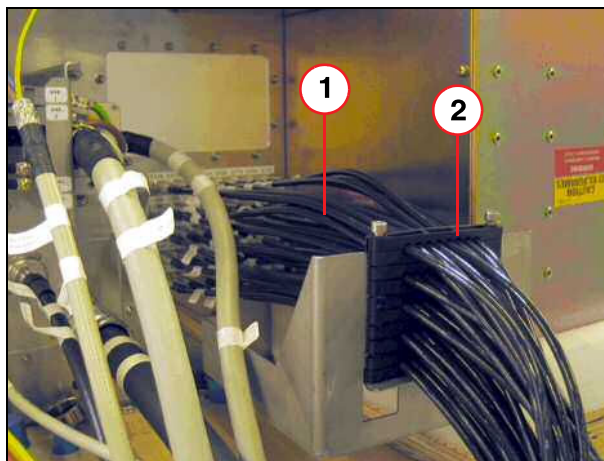
Fig. 13: Magnet



- (1) RF cable
- (2) Strain relief

- Route the cables at the magnet the same way as the cables previously installed.
- Connect the RF cables (1) at the magnet.
- Secure them using strain reliefs (2).
- Route the cables to the filter plate.

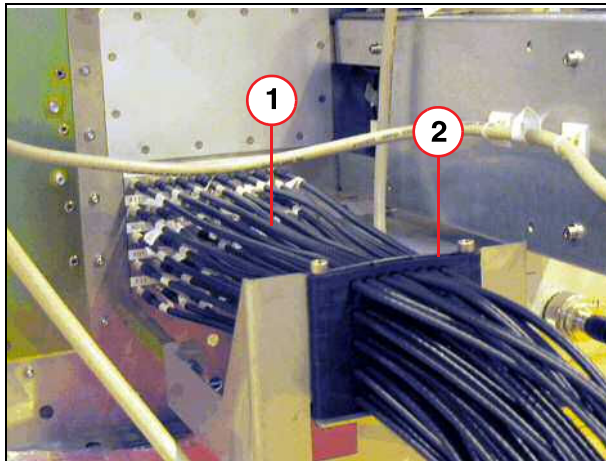
Fig. 14: Filter plate RF room



- (1) RF cable
- (2) Strain relief

- Connect the RF cables (1) at the filter plate in the RF room.
- Secure them using strain reliefs (2).

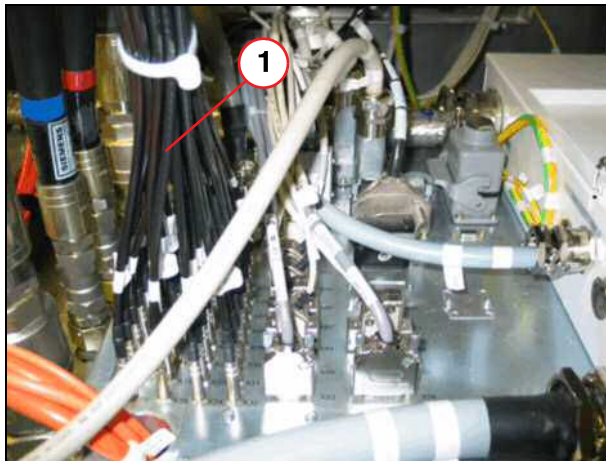
Fig. 15: Filter plate, electronics room



- (1) RF cable
- (2) Strain relief

- Route the cables from the filter plate to the ACC cabinet.
- Connect the RF cables (1) at the filter plate in the electronics room.
- Secure them using strain reliefs (2).

Fig. 16: ACC cabinet



- (1) RF cable

- Connect the RF cables (1) on top of the ACC cabinet.

2.2 ACC Upgrade

Tools:

- 2 mm Allen key (for removing the blind plates and after inserting the PCI-TX boards)
- Torx key TX8 (for removing the blind plates and after inserting the REC assemblies)



CAUTION

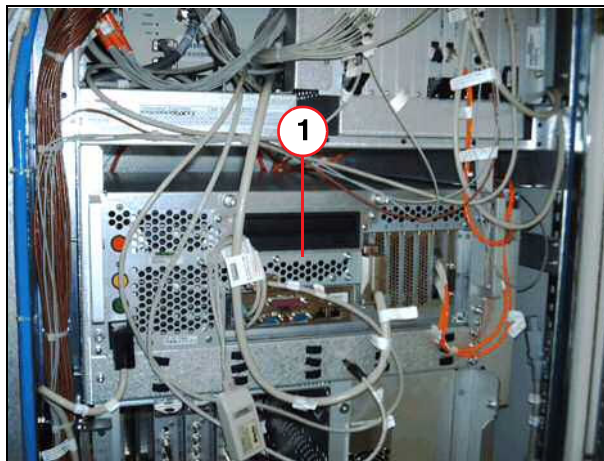
The Lamda power supply on the inside back wall of the Imager is always powered on, even when the system is switched off.

» Hazardous conditions. Do not touch the power supply unit.



The Imager weighs approximately 20 kg.

Fig. 17: ACC cabinet/Imager



(1) Imager

- Removing the cables from the Imager (1).
- Remove the Imager.

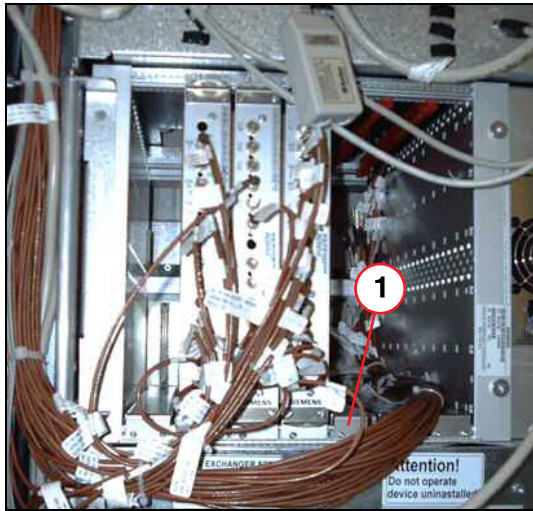


Remove only the blind plates of the RFSU that will be replaced by an REC assembly (fill to the right, beginning with the receiver cassette).

When removing a blind plate without replacing it by a REC assembly, the cooling of the REC assemblies is not ensured.

A blind plate has to be present for plug-in-locations that are not required.

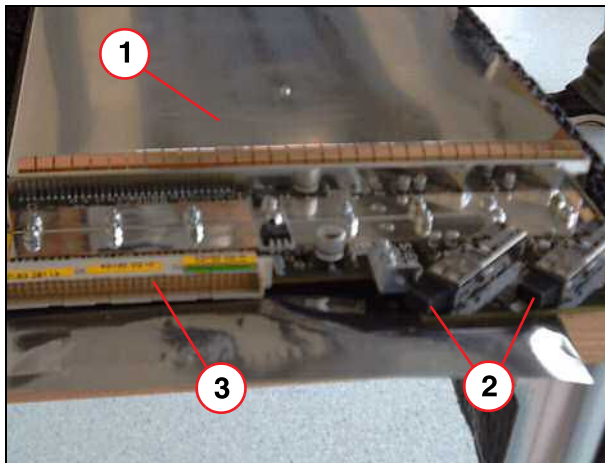
Fig. 18: RFSU/Blind plates



(1) Blind plates

- Remove the blind plates (1) of the RFSU.

Fig. 19: REC assembly



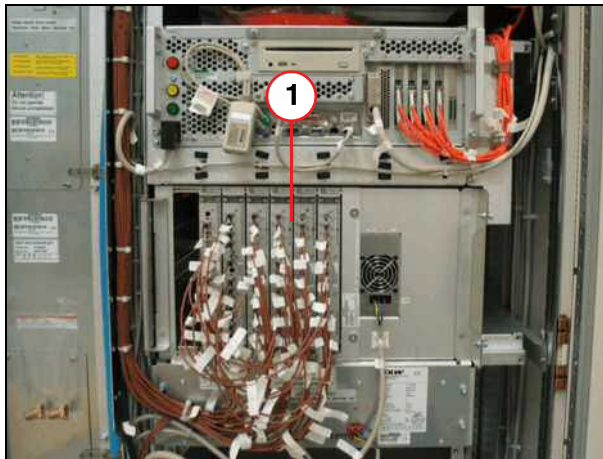
(1) REC assembly
 (2) Dust protection
 (3) Terminal connector

- Unpack the REC assemblies (1)
- Remove the dust protection (2) of the REC assemblies.



When you swing out the rack, ensure that the cables and fiber optic cables are not damaged.

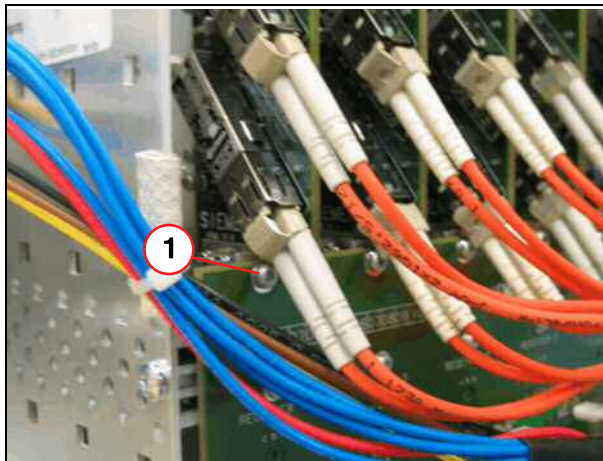
Fig. 20: RFSU/Receiver



(1) Receiver

- Slide in the receiver (1). Ensure that the terminal connector (→ 3/ Fig. 19 Page 18) on the back plate is okay.
- Swing out the rack.

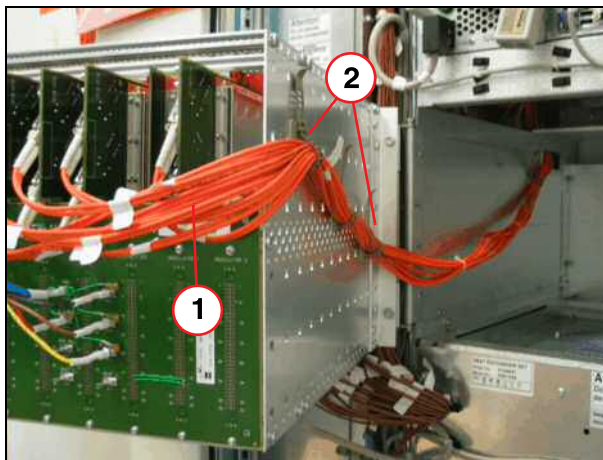
Fig. 21: Back of RFSU



(1) Screw

- Attach the receiver to the back plane (1) using screws.

Fig. 22: RFSU/Fiber optic cable routing



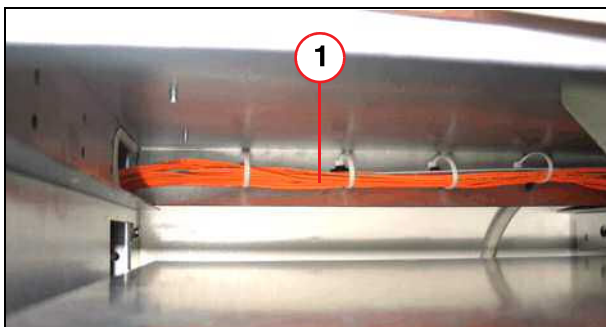
(1) FOC (Fiber optic cable)
(2) Cable tie

- Insert fiber optic cable (1) and route it from the RFSU to the Imager.
- Route the fiber optic cable in the already existing cable ties that can be locked again (2).
- If required, secure the fiber optic cables using additional cable ties.

Caution: Do not crush the FOC with the cable ties.

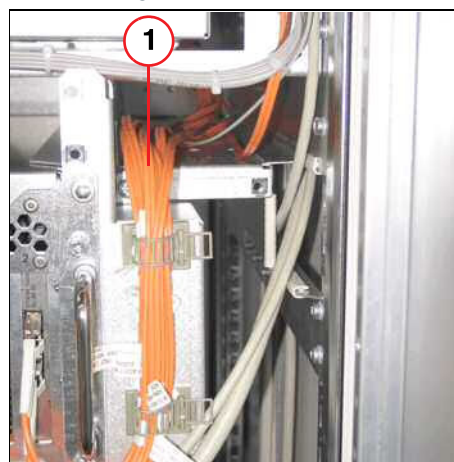
- Route the fiber optic cable (1) in the ACC cabinet and attach it with cable ties.

Fig. 23: FOC routing (ACC inside, above the imager)



(1) FOC

Fig. 24: FOC routing (ACC inside, along side the imager)



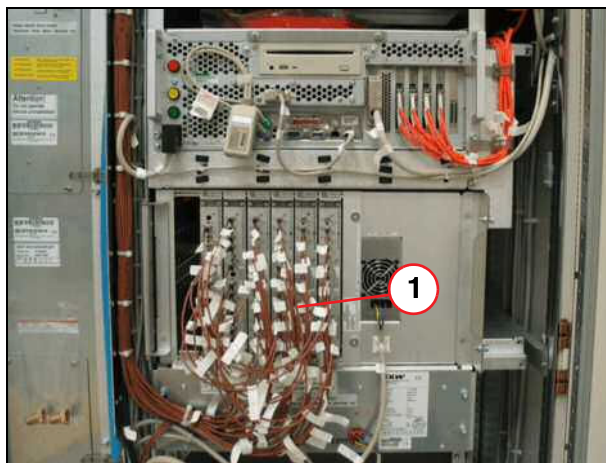
(1) FOC

- Close the rack.



When you swing out the rack, ensure that the cables and fiber optic cables are not damaged.

Fig. 25: RFSU/RF cable



(1) RF cable

- Insert RF cable (1) in the front of the RFSU (top of REC, MOD-REC, or synthesizer REC) and attach with cable ties.

Caution: Do not crush the cable with the cable ties.



Attach the RF cables with the cable ties in a way that allows opening the swivel rack.

2.2.1 Imager



There are various configurations for the imager, as described in the following:

1. (→ Imager MR005 Supermicro / Page 21)
or
2. (→ Imager MR005 Dual Opteron / Page 22)
or
3. (→ MCPPro Multi 2 (Dual Opteron with PCI_RX 16 boards) / Page 22)
or
4. (→ High-end imager (MCPPro Multi 8) / Page 23)



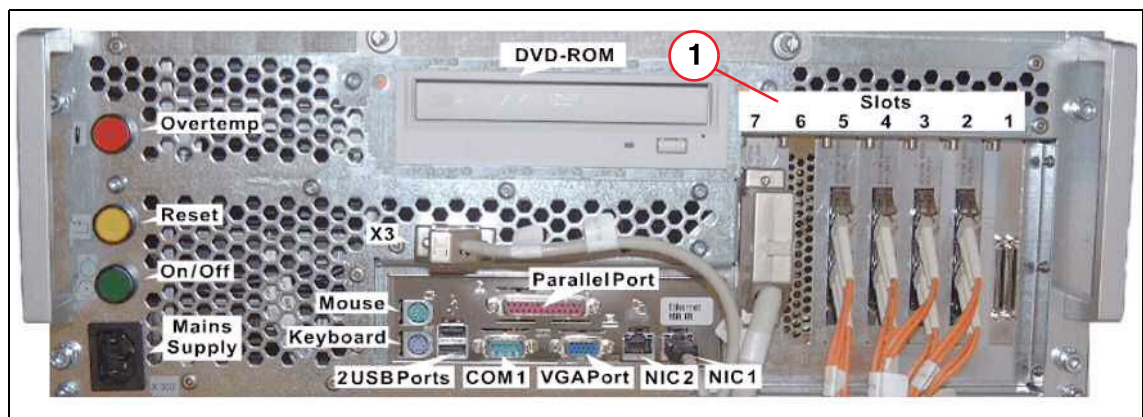
The slots are configured differently for Imager MR005 Supermicro and Imager MR005 Dual Opteron (however, the designations for the slots are identical)

2.2.1.1 Imager MR005 Supermicro

The PCI_RX boards included in the delivery volume have to be installed for the MR005 Supermicro.

- Install the PCI_RX board included in the delivery volume in the imager; refer to (→ Installing the PCI-RX boards (optional) / Page 24)

Fig. 26: Front view of Imager MR005 Supermicro



(1) Slots

Tab. 1 Imager MR005 Supermicro Slots

Label	Use
Slot 1	Adaptec RAID controller
Slot 2	PCI_RX1 board
Slot 3	PCI_RX2 board (optional)
Slot 4	PCI_RX3 board (optional)
Slot 5	PCI_RX4 board (optional)

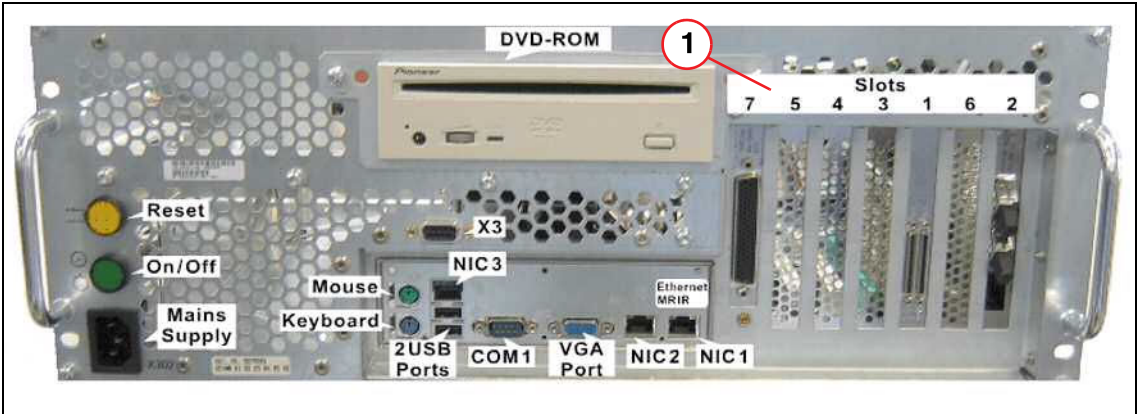
Label	Use
Slot 6	Not used
Slot 7	Signal Distributor

2.2.1.2 Imager MR005 Dual Opteron

The PCI_RX boards included in the delivery volume have to be installed for the MR005 Supermicro.

- Install the PCI_RX board included in the delivery volume in the imager; refer to (→ Installing the PCI-RX boards (optional) / Page 24)

Fig. 27: Front view of Imager MR005 Dual Opteron



(1) Slots

Tab. 2 Imager MR005 Dual Opteron Slots

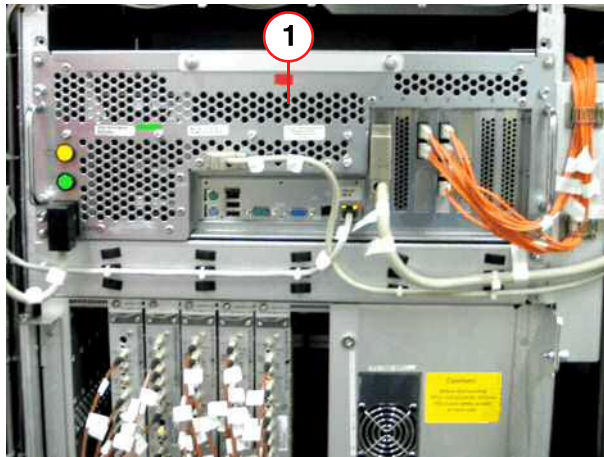
Label	Use
Slot 2	PCI_RX1 board
Slot 6	Not used
Slot 1	Adaptec RAID controller
Slot 3	PCI_RX2 board (optional)
Slot 4	PCI_RX3 board (optional)
Slot 5	PCI_RX4 board (optional)
Slot 7	Signal Distributor

2.2.1.3 MCPro Multi 2 (Dual Opteron with PCI_RX 16 boards)



The new MCPro Multi 2 already includes the PCI-RX boards (depending on the upgrade level).

Fig. 28: Imager MCPPro Multi 2



(1) MCPPro Multi 2

- Install the new MCPPro Multi 2 (1).
- Connect fiber optic cable (2) and attach with cable ties.
- Connect the cables.

Tab. 3 Imager MR005 Dual Opteron Slots with PCI_RX 16 boards

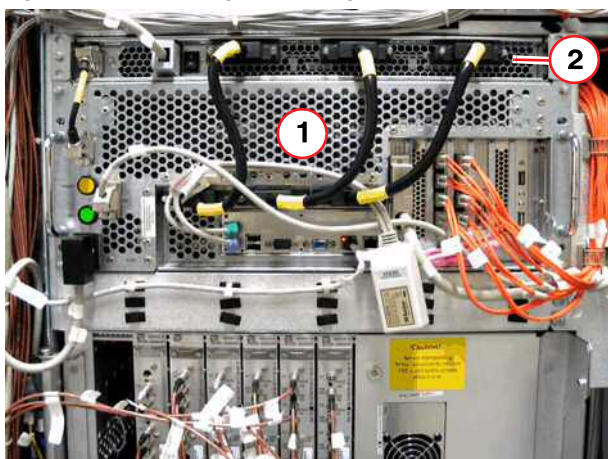
Label	Use
Slot 5	Not used
Slot 6	Not used
Slot 1	Adaptec RAID controller
Slot 2	PCI_RX2 16 board 1
Slot 3	PCI_RX3 16 board 2 (optional)
Slot 4	Not used
Slot 7	Signal Distributor

2.2.1.4 High-end imager (MCPPro Multi 8)



The new high-end imager already includes the PCI-RX boards (depending on the upgrade level).

Fig. 29: ACC with high-end imager



- (1) High-end CPU
- (2) Hard disk module

- Install the new high-end imager (1).
- Connect fiber optic cable (2) and attach with cable ties.
- Connect the cables.



Use the following document to install the high-end imager; refer to (→ High-end Imager (MCPPro Multi 8) / MR00-000.814.30) in the "Technical Documents, Volume 2", chapter Options.

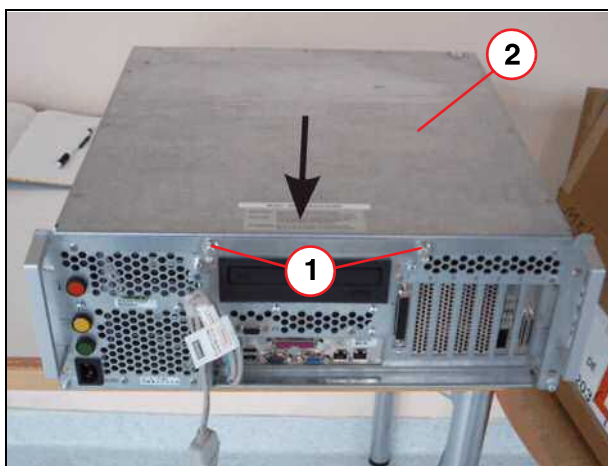
2.2.1.5 Installing the PCI-RX boards (optional)



The PCI_RX boards are only installed for the MR005 Supermicro and the MR005 Dual Opteron.

For the MR005 Dual Opteron with PCI_RX 16 boards and high end, the imager is replaced.

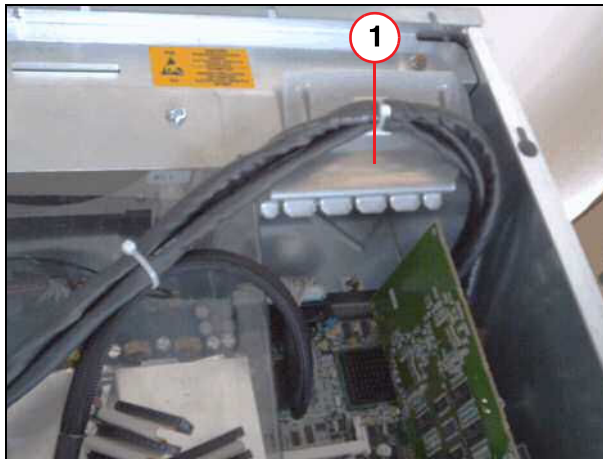
Fig. 30: Imager



- (1) Cover attachment
- (2) Cover

- Loosen the cover mounting (1) and cover (2) and slide toward the front.

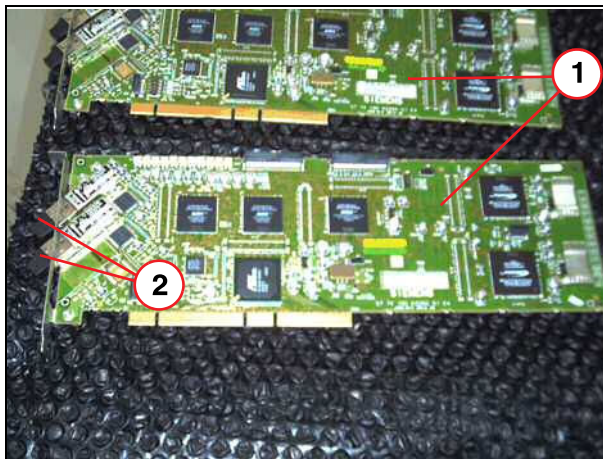
Fig. 31: Imager/Holder



(1) Holder

- Loosen the holder (1) for the boards and slide it upward.

Fig. 32: Imager/Boards



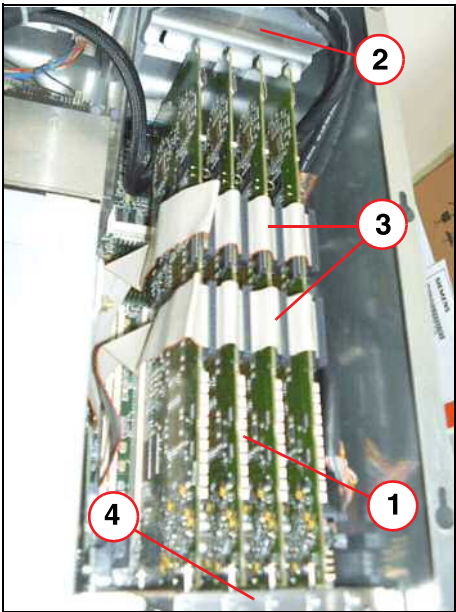
(1) Boards (PCI-RX boards)
(2) Dust protection

- Unpack the boards (1).
Jumpering is not necessary.
- Remove dust cover (2).



When connecting the ribbon cable, ensure that any excessive length between the boards is routed above using a loop hanger.

Fig. 33: Imager/Inserting Boards



- (1) Boards (PCI-RX boards)
- (2) Holder
- (3) Ribbon cables
- (4) Blind plates

- Remove blind plates (4) (only those required for inserting the boards)
- Insert the boards (1).
Configuration, refer to
(→ Tab. 4 Page 26) or
(→ Tab. 5 Page 26)
- Reattach holder (2) for the boards.
- Insert ribbon cables (3).
- Attach cover and lock attachments into place.

Tab. 4 Imager MR005 Supermicro Slots

Label	Use
Slot 1	Adaptec RAID controller
Slot 2	PCI_RX1 board
Slot 3	PCI_RX2 board (optional)
Slot 4	PCI_RX3 board (optional)
Slot 5	PCI_RX4 board (optional)
Slot 6	Not used
Slot 7	Signal Distributor

Tab. 5 Imager MR005 Dual Opteron Slots

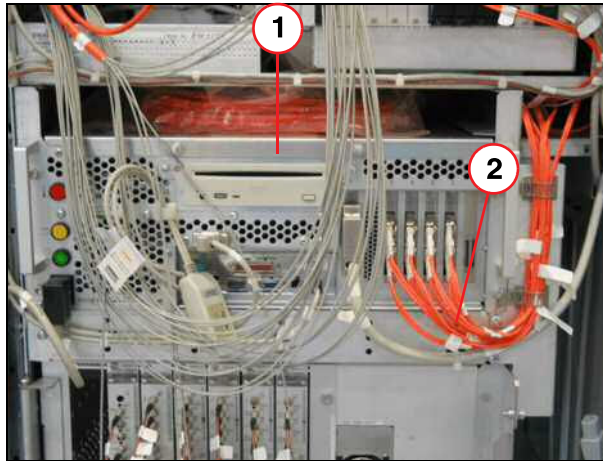
Label	Use
Slot 2	PCI_RX1 board
Slot 6	Not used
Slot 1	Adaptec RAID controller
Slot 3	PCI_RX2 board (optional)
Slot 4	PCI_RX3 board (optional)

Label	Use
Slot 5	PCI_RX4 board (optional)
Slot 7	Signal Distributor



Ensure that none of the cables is damaged when installing the Imager.

Fig. 34: Installing the Imager



- (1) Imager
- (2) FOC (Fiber optic cable)

- Installing the Imager (1).
- Connect fiber optic cable (2) and attach with cable ties.
- Connect the cables.

2.2.2 Return

Place the replaced components (Imager) in the packaging from the new components and return to the following address:

2.2.2.1 European Union (EU)

Siemens AG Healthcare Sector
CO ME
Geis Industrieservice GmbH
An den Sandgruben 1
D-91058 Forchheim
Germany

2.2.2.2 From outside the EU:

Siemens AG Healthcare Sector

Logistics Service Center
World Distribution Center
c/o Geis Industrie Service GmbH
An der Gehespitz 90
D-63263 Neu-Isenburg
Germany

3.1 Software

- Login as meduser
- At the "MRC", simultaneously press **Ctrl - Esc**. Select **Start, Advanced User**; enter the advanced user password; click **OK**.

3.1.1 Values that must be read out manually (Espree Z - DZ Engine only)

Read the following values from the software

- Open a Command Prompt
- Enter **modifymeaspara** ↵
- Select item **2** ↵ (gradient offsets) from the menu.
- Note the values for "G_x", "G_y", and "G_z" in mT/m
- Select item **3** ↵ (SuperConductiveShimCurrent) from the menu.
- Note the value for "superConductivityShimCurrent" in mA
- End the program.

3.1.2 Installing the licenses

- Starting the local service interface **Configuration, Service, Licensing** in the field "Import license file from local drive:". Navigate with **Browse** to locate the licensing file on the medium provided with the delivery.

Licenses				
Avanto	Espree	Symphony A Tim System	Trio A Tim System	Verio
■ Tim [76x18]	■ Tim [76x18]	■ Tim [76x18]	■ Tim [102x18]	■ Tim [102x18]
■ Tim [76x32]	■ Tim [76x32]	■ Tim [76x32]	■ Tim [102x32]	■ Tim [102x32]
■ SQ-engine:	■ DZ engine ¹	n.a.	n.a.	n.a.

1. The **Slim-line gradient coil** is the prerequisite for the upgrade from the Z to the DZ engine.

3.1.3 Imager

For system < **VB13**:

- Installing the software.

To install the software, the following steps have to be performed under "Installation Software", "Installation Software at the Imager".

For systems = **VB13**:

- Navigate in the file explorer on the MRC to folder c:\MRIR\Med\config\scripts, delete the file setup.lastcall.

- **Reboot** the system.

For systems > **VB13**:

- Navigate in the file explorer on the MRC to folder c:\Medcom\MCIR\Med\config\scripts, delete the file setup.lastcall.
- **Reboot** the system.

To install the software, the following steps have to be performed under "Installation Software", "Installation Software at the Imager" (applies to systems < VB13 only).

3.1.4 Values that must be entered manually (Espree Z - DZ Engine only)

Enter the following values into the software

- Open a Command Prompt
- Enter **modifymeaspara** ↵
- Select item **2** ↵ (gradient offsets) from the menu.
- Enter the values for "G_x", "G_y", and "G_z" in mT/m
- Select item **3** ↵ (SuperConductiveShimCurrent) from the menu.
- Enter the value for "superConductivityShimCurrent" in mA
- End the program.

3.1.5 System configuration

3.1.5.1 RF upgrade

- Start the local service interface **Configuration, Service, Measurement, Meas. Settings**. Go to the **pull-down menu** "Number of Receiver Channels" and select the appropriate number of channels.

Upgrade variations	from 8 to 18	from 18 to 32	from 8 to 32
18	X		
32		X	X

- Perform the following steps: **Save, Finish, OK**, and reboot the system.

3.1.5.2 GPA Upgrade (for Avanto and Espree only)

- Start the local service interface **Configuration, Service, Measurement, Meas. Settings**. Go to the **pull-down menu** "Gradient Power Amplifier" and select the relevant gradient amplifier.

Upgrade variations	Avanto		Espree	
	Q	SQ	Z	DZ
K2259/1500V/550A	X		X	
K2259/2000V/650A		X		X

- Perform the following steps: **Save, Finish, OK**, and reboot the system.

3.2 Tune-up

- Start the local service interface **Tune-Up**
- Perform the **Tune-Up** steps (the steps to be performed are automatically activated).
Perform the following additional steps:
 - Relative Receive Path Calibration
 - Absolute Receiver Path Calibration
 - RF characteristic und subsequent tests (for Verio systems because of the BC pickup coil connectors)

3.3 Testing pick-up coils/Body Coil



This test is part of the Quality Assurance (QA) with software version N4 VB15A and N4 VD13A.



This test is valid for all systems with software version N4 VB13.
Except for the Espree Straight Bore system and the Tim system with 8 Receiver Channels configuration.

Auxiliary tools:

- Loader with 24 cm spherical phantom

Preliminary steps:

- Register a patient (any)
- Move the table to HOME position
- Coil and phantom selection:
 - Remove all local transmit coils
 - Remove all local receiver coils
 - Position the loader together with the 24 cm spherical phantom (centered in the loader) in the upper part of the table and move it under the light localizer.
- Switch the light localizer on and off.
- Move the table into the CENTER position.

3.3.1 Performing the measurement

- Start the measurement with one of the protocols (e.g. a localizer).
 - » Measurement can be stopped immediately after it begins.

3.3.2 Evaluation

- Open the file "RFSWDHistoryListNew.log"
(it is located in c:\MedCom\Log Directory)
- Scroll back from the last entry in the file to the "604 DiCoTest Tra Power[0]" column
- Once there, ensure that the entries for the following parameters reach a minimum of 200 V or maximum 2500 V
 - 2500 > DiCoTest PickUp Probe Ampl. [0][0] > 200 V
 - 2500 > DiCoTest PickUp Probe Ampl. [0][1] > 200 V

This concludes the test.

- » If the conditions are not met, check the connections from the output BC to the input receiver.

3.4 Tests to be performed

RF upgrade	▪ MOD/REC Test ▪ Functional RCCS test
GPA Upgrade ¹	▪ GPA tests

1. Only for the GPA Upgrade Avanto and Espree

3.5 QA

- Start the local service interface **QA**
- Perform the **QA** steps (the steps to be performed are automatically activated).
The following step needs to be performed in addition:
 - **SAR monitor test** (for Verio systems because of the BC pickup coil connectors).

Revision 02

- Routing the RF cables in the ACC cabinet (→ Installing the RF cable in the ACC / Page 10).
- Different configurations of the slots for Imager (→ Imager MR005 Supermicro / Page 21) and (→ Imager MR005 Dual Opteron / Page 22).
- Adjusted for Trio, A Tim System and Symphony, A Tim System.
- Editorial changes.

Revision 03

- Adapted for the Espree system.
- Editorial changes.

Revision 04

- (→ Fig. 11 Page 14) added.
- Editorial changes.

Revision 05

- Path for systems > VB13 changed.
- Verio system added.
- Editorial changes.

Revision 06

- (→ BC pickup coil connectors (only for Verio) / Page 9) added.
- Tuneup und QA changed.

Revision 07

- Added a note about the receiver (→ Delivery volume / Page 7)

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